

Computer vision in deep learning implementation and its applications

YOLO11

YOLO11介紹(Ultralytics YOLO11)

- Ultralytics 是一家專注於電腦視覺 (computer vision) 解決方案的公司，標榜「開放原始碼 + 商業化」並行。
- 其使命之一是「賦能人們與公司釋放 AI 的正面潛能」。
- GitHub 上其開放原始碼 YOLO 框架（如 YOLOv5、YOLOv8、YOLO11 等）備受社群關注。
- 公司總部 / 註冊在美國，但亦有全球布局。

[Products](#)[Solutions](#)[Resources](#)[Licensing](#)[Company](#)[GitHub](#) ★ 119,052[Get in touch](#)

EN



Blues
Dog Executive Officer



Glenn Jocher
Founder & CEO



Paula Derrenger
VP of Growth



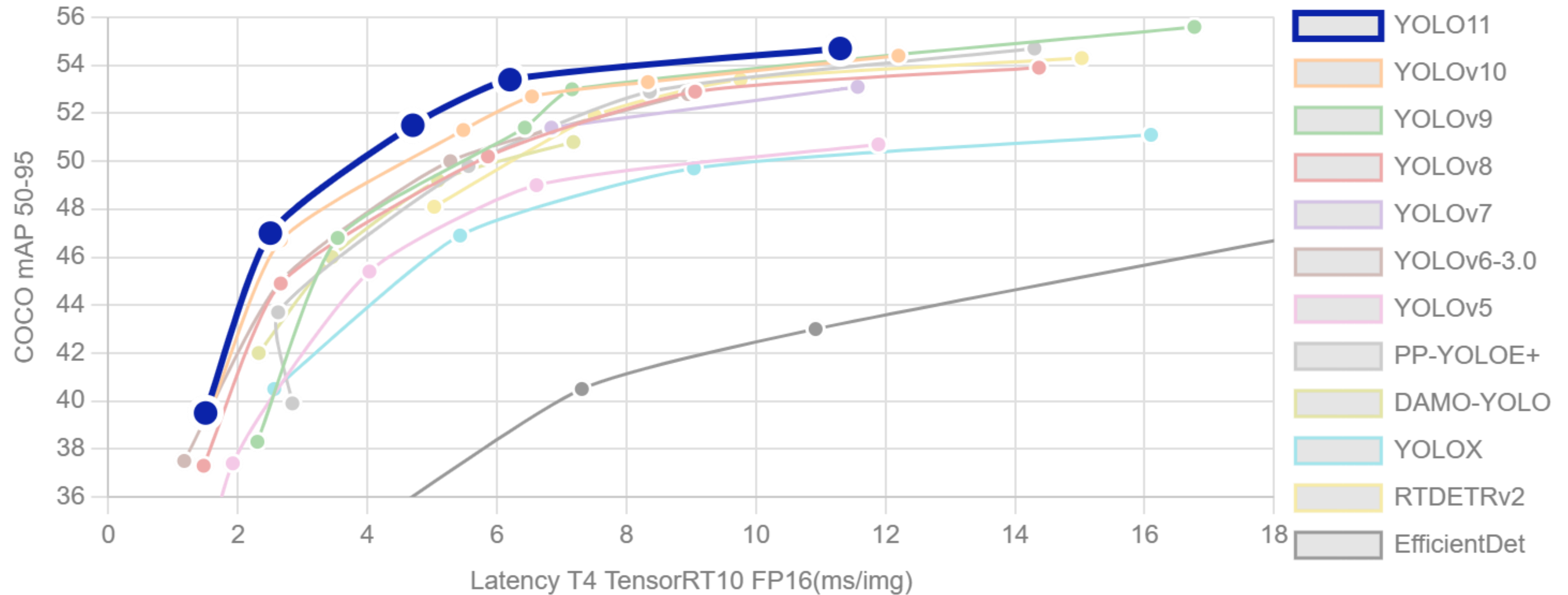
William Beaven
Head of Legal



Sergiu Waxmann
Lead Web Developer

YOLO11介紹(Ultralytics YOLO11)

YOLO11 is the latest iteration in the Ultralytics YOLO series of real-time object detectors, redefining what's possible with cutting-edge accuracy, speed, and efficiency.



<https://docs.ultralytics.com/models/yolo11/>

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Performance

Detection (COCO)

Segmentation (COCO)

Classification (ImageNet)

Pose (COCO)

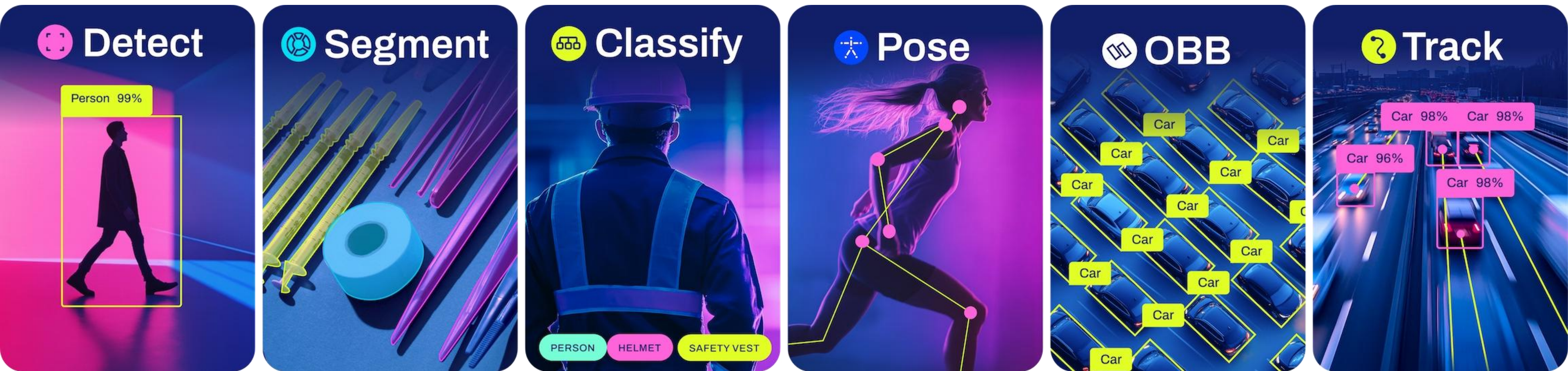
OBB (DOTAv1)

See [Detection Docs](#) for usage examples with these models trained on [COCO](#), which include 80 pre-trained classes.

Model	size (pixels)	mAP ^{val} ₅₀₋₉₅	Speed CPU ONNX (ms)	Speed T4 TensorRT10 (ms)	params (M) $\downarrow$	FLOPs (B)
YOLO11n	640	39.5	56.1 $\pm$ 0.8	1.5 $\pm$ 0.0	2.6	6.5
YOLO11s	640	47.0	90.0 $\pm$ 1.2	2.5 $\pm$ 0.0	9.4	21.5
YOLO11m	640	51.5	183.2 $\pm$ 2.0	4.7 $\pm$ 0.1	20.1	68.0
YOLO11l	640	53.4	238.6 $\pm$ 1.4	6.2 $\pm$ 0.1	25.3	86.9
YOLO11x	640	54.7	462.8 $\pm$ 6.7	11.3 $\pm$ 0.2	56.9	194.9

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YOLO11介紹(Ultralytics YOLO11)

Models

YOLOv6

YOLOv7

YOLOv8

YOLOv9

YOLOv10

YOLO11

YOLO12

YOLO26 🚀 Coming soon

YOLO12: Attention-Centric Object Detection



Overview

YOLO12 introduces an attention-centric architecture that departs from the traditional CNN-based approaches used in previous YOLO models, yet retains the real-time inference speed essential for many applications. This model achieves state-of-the-art object detection accuracy through novel methodological innovations in attention mechanisms and overall network architecture, while maintaining real-time performance. Despite those advantages, YOLO12 remains a community-driven release that may exhibit training instability, elevated memory consumption, and slower CPU throughput due to its heavy attention blocks, so Ultralytics still recommends YOLO11 for most production workloads.

Community Model

YOLO12 is maintained primarily for benchmarking and research. If you need stable training, predictable memory usage, and optimized CPU inference, choose [YOLO11](#) or other Ultralytics-maintained releases for deployment.

<https://docs.ultralytics.com/models/yolo11/>

YOLO11介紹(Ultralytics YOLO11)

Models
YOLOv6
YOLOv7
YOLOv8
YOLOv9
YOLOv10
YOLO11
YOLO12
YOLO26 🚀 Coming soon

Ultralytics YOLO26

 **Coming Soon** ⚠️

🚧 YOLO26 models are still under development and not yet released. Performance numbers shown here are **previews only**. Final downloads and releases will follow soon — stay updated via [YOLO Vision 2025](#).

YOLO11介紹(Ultralytics YOLO11)

Ultralytics has not published a formal research paper for YOLO11

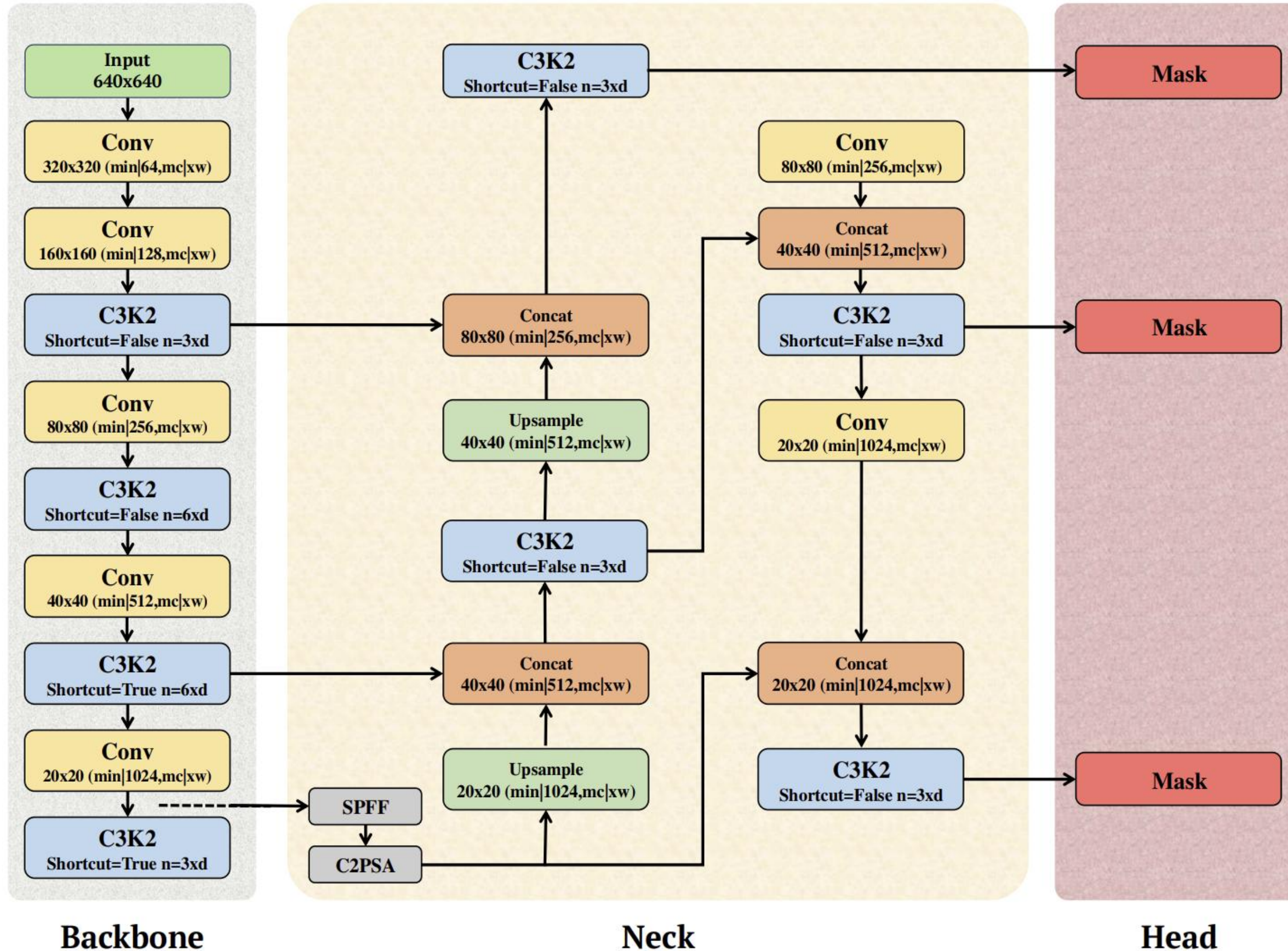
Citations and Acknowledgments



Ultralytics YOLO11 Publication

Ultralytics has not published a formal research paper for YOLO11 due to the rapidly evolving nature of the models. We focus on advancing the technology and making it easier to use, rather than producing static documentation. For the most up-to-date information on YOLO architecture, features, and usage, please refer to our [GitHub repository](#) and [documentation](#).

YOLO11介紹(Ultralytics YOLO11)



- Backbone：影像特徵萃取
- Neck：多尺度特徵融合
- Head：偵測與遮罩輸出

YOLO11範例程式

<https://colab.research.google.com/drive/1SyQzzRkHxHiJcCN2Bylo19NzHXXi5B3V?usp=sharing>

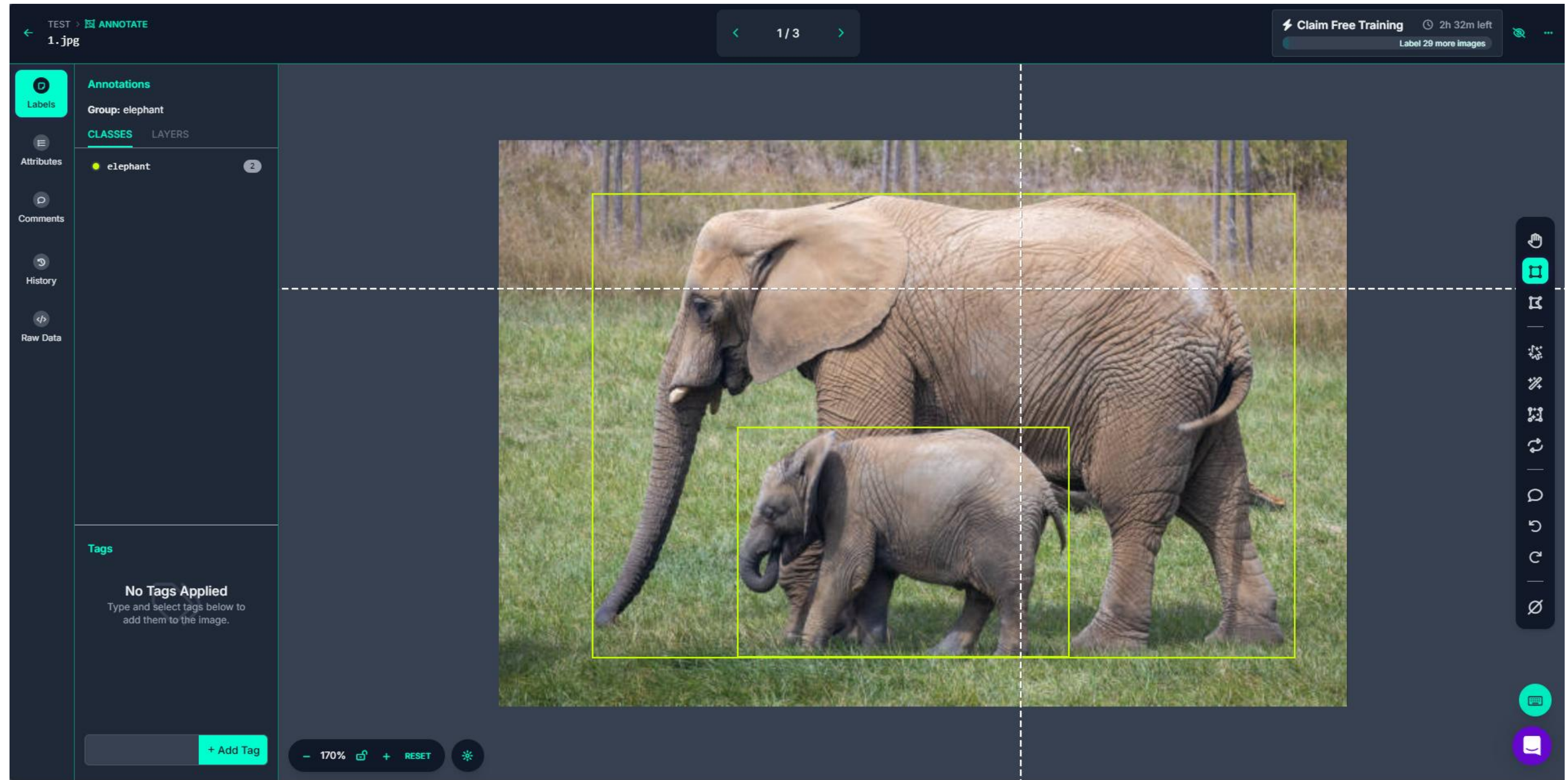
資料標註平台



- Roboflow 是一個專注於電腦視覺（ Computer Vision ）流程工具與服務的平台。
- 成立時間為 2019 年，由 Brad Dwyer 與 Joseph Nelson 創辦，總部設於美國愛荷華州得梅因（ Des Moines, Iowa ）
- 它整合了一套「從資料集管理 → 標註 → 模型訓練 → 部署」的流程，使得視覺模型開發對使用者更為友善。

<https://roboflow.com/about>

資料標註平台



資料標註平台

Roboflow 標註 → Export 成 Ultralytics YOLO / YOLOv5 格式 →
拿裡面的 data.yaml + images/ + labels/ → 直接丟給 YOLO11 yolo train 使用

```
results = model.train(data="coco8.yaml", epochs=3, imgsz=640)
```

<https://roboflow.com/about>

YAML(/'jæməɪl/)是什麼？

YAML (YAML Ain't Markup Language) 是一種：

- 人類可讀性高
- 簡潔又容易編寫
- 常用在設定檔、資料格式交換

的純文字格式。

YOLO 資料集 YAML

```
ultralytics/cfg/datasets/coco8.yaml

# Ultralytics 🚀 AGPL-3.0 License - https://ultralytics.com/license

# COCO8 dataset (first 8 images from COCO train2017) by Ultralytics
# Documentation: https://docs.ultralytics.com/datasets/detect/coco8/
# Example usage: yolo train data=coco8.yaml
# parent
# └─ ultralytics
#   └─ datasets
#     └─ coco8 ← downloads here (1 MB)

# Train/val/test sets as 1) dir: path/to/imgs, 2) file: path/to/imgs.txt, or 3) list: [path/to/imgs1, pa
path: coco8 # dataset root dir
train: images/train # train images (relative to 'path') 4 images
val: images/val # val images (relative to 'path') 4 images
test: # test images (optional)

# Classes
names:
  0: person
  1: bicycle
  2: car
  3: motorcycle
  4: airplane
  5: bus
  6: train
```


YOLO 標註格式 (.txt)

每行代表一個物件，內含：class_id x_center y_center width height，全部以空白 (space) 分隔。

1. class_id (類別編號) 從 0 開始編號例如：
0 : person
1 : car
2 : dog
類別對應在 data.yaml 中的 names: 區塊
2. x_center, y_center (邊界框中心點座標)
 - 表示 bounding box 的中心相對座標 (0~1)
3. width, height (邊界框寬高)
 - 相對座標 (0~1)

假設一張圖片裡有兩個物件：

```
0 0.521 0.633 0.212 0.301
```

代表：

```
2 0.272 0.400 0.155 0.210
```

•第 1 行：

類別 0，框中心 (0.521, 0.633)，大小 (0.212 × 0.301)

•第 2 行：

類別 2，框中心 (0.272, 0.400)，大小 (0.155 × 0.210)

YOLO 標註格式 (.txt)

4. 每張圖片對應 1 個同名 .txt 標註檔

images/train/0001.jpg

labels/train/0001.txt

images/train/0002.jpg

labels/train/0002.txt

images/train/0003.jpg

labels/train/0003.txt

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5. YOLO 的標註不包含：

- 不包含物件名稱（只能用 class_id）
- 不包含絕對 pixel 值（只能用 0~1 normalized）
- 不包含 polygon、多邊形（除非你使用 segmentation 模式）

Q & A

期中Project

1. 專案目的 Project Objectives

- 能夠自行尋找與分析影像資料集
- 能夠實作深度學習影像分類 / 偵測 / 分割模型
- 能針對資料特性進行前處理與資料增強
- 能夠視覺化模型結果，進行錯誤分析
- 能夠撰寫報告與口頭簡報呈現實驗與結論

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2. 專案題目 Project Theme

Theme1 : Rock Paper Scissors Dataset (Image Classification)

<https://www.kaggle.com/datasets/sanikamal/rock-paper-scissors-dataset/data>

Theme2 : 請至 Kaggle 選擇任一影像相關資料集可包含以下任一主題：

- Image Classification
- Object Detection
- Instance / Semantic Segmentation
- ...

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3. 專案流程 Project Workflow

(1) Dataset 選擇與整理

- 介紹資料集來源、用途、影像尺寸 / 格式
- 訓練 / 驗證 / 測試切分方式
- 必要的資料清洗 (移除噪音、標籤錯誤...)

(2) 資料前處理與增強

- normalization / resizing
- rotation / flip / crop / color jitter
- 其他適合的 augmentation

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3. 專案流程 Project Workflow

(3) 模型建立

可使用：

- CNN baseline (ResNet / EfficientNet...) 、物件偵測 (YOLOv8/11, Faster R-CNN) 、 Segmentation (UNet, DeepLabV3) 、 Transformer

需說明：

- 模型架構
- 超參數 (learning rate, epoch, batch size)
- 損失函數、優化器

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3. 專案流程 Project Workflow

(4) 訓練與評估

- 訓練曲線：loss / accuracy (或 mAP / IoU)
- confusion matrix
- error analysis (錯誤案例分析)
- ablation study (消融實驗) (optional)

(5) 結果視覺化與討論

- 模型最佳示例、錯誤示例
- 與 Kaggle上baseline比較(比較方式及對象不限)

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4. 需提交項目

(1)簡報檔(螢幕錄影10-12分鐘報告)，需包含：

- 資料集介紹
- 模型架構與訓練方式
- 實驗結果（清楚圖表）
- 模型表現分析
- 結論與未來改善

(2)原始碼(**ipynb檔**，檔名為：姓名_學號_Midterm Project)

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5. 評分：Theme1(30%)、Theme2(70%)，依流程完整性評分。

6. 繳交期限：2025/12/12 中午12:00(11/28實作周停課一次)